



A LITTLE DOMESTIC BUREAUCRACY

Set ENERGY. Set TOLERANCE. Press NEGOTIATE. Receive a politely over-official suggestion for what to do next.

Walnut case, two ash dials, brushed aluminium and a quiet paper-like display. No sound, microphone, Wi-Fi or subscription. The object reads the controls; it does not read people.

The first response library contains 27 original lines across nine settings. Change the humour in a local text file without remaking the woodwork.

This tests a different kind of value: a useful little ritual that can remain pleasant on a shelf. Customer appeal and retail economics still need evidence.

Body 8 W x 5.5 D x 2.25 H, excluding feet and controls. Flat tabletop prototype. The separate faceplate lets us change the display or controls while retaining the walnut body.

FINISHED PARTS

Lid: 8 x 5.5 x 0.500 walnut.

Lid opening: X/Y0.750, 6.500 x 4.000, R0.125.

Two rings: 8 x 5.5 x 0.750, minimum rim 0.500.

Base: 8 x 5.5 x 0.250 actual birch plywood.

Faceplate: 7 x 4.5 x 1.5 mm aluminium.

Knobs: diameter 1.375 x 0.625; qty 2 plus coupon.

Window: 3.125 x 1.625 x 0.0625 nominal clear plastic.

Display carrier blank: 4.250 x 2.250 x 0.125.

Base top Z0.250; lid underside Z1.750; lid top Z2.250. Surface plate top Z2.3091. Full underside opens for service. Solid parts share grain direction; slotted plywood fixings allow movement.

Nine 2D inch DXFs. Control bores, display mounts and cable notch await measured samples. This is not a STEP model, toolpath or nesting layout. Check units on import and let the operators choose cutters and workholding.

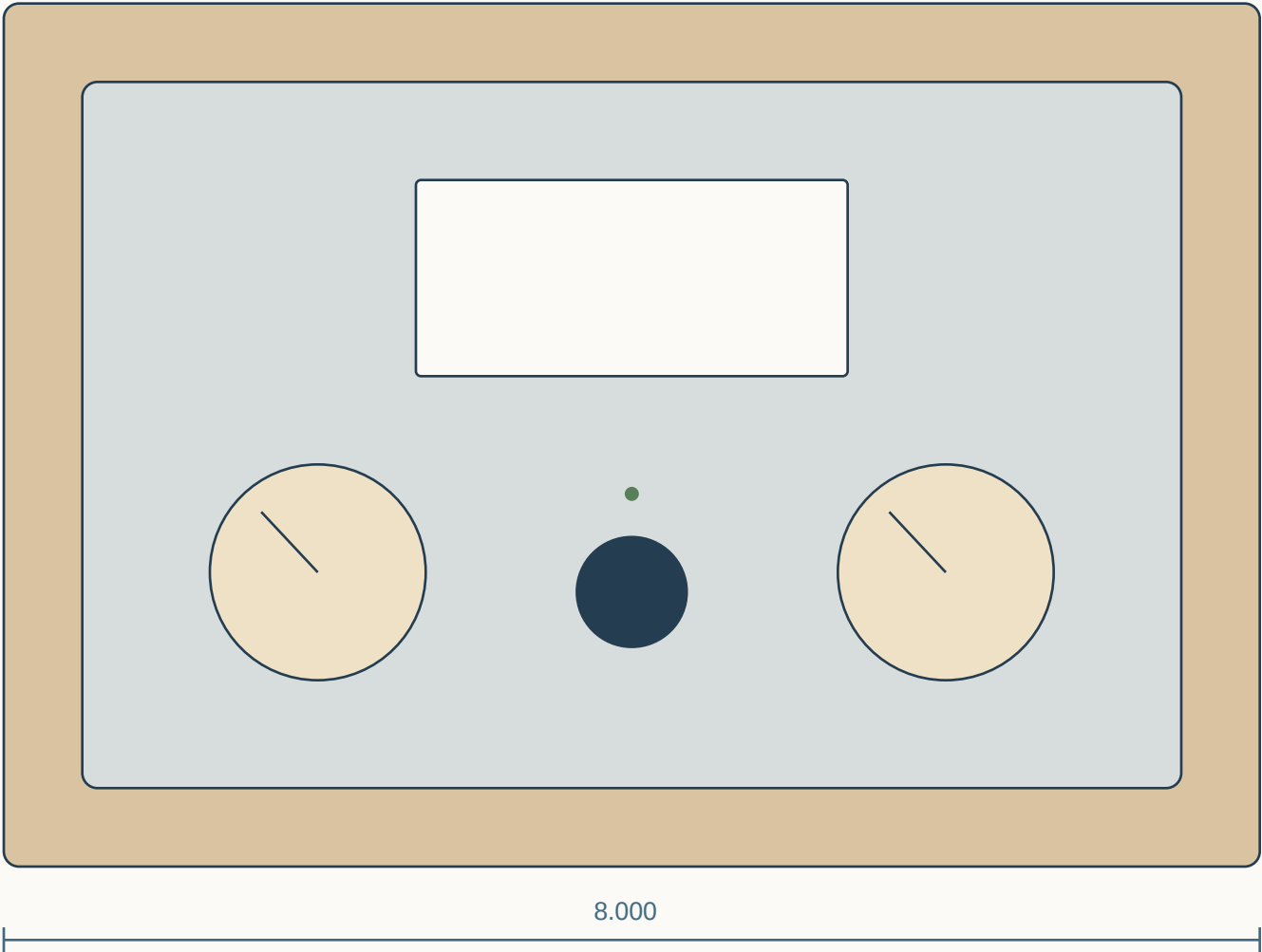
Cut a knob coupon and a display-window coupon first. Small parts require tabs or fixtures; nominal 4/4 stock does not guarantee finished yield.



PLAN: ring, qty 2. Outer R0.250; grain along X.

SIDE STACK / Z FROM BASE UNDERSIDE

|                     |
|---------------------|
| LID .500            |
| TWO RINGS .750 EACH |
| BASE .250           |



GLOBAL LOCATIONS / FACEPLATE OFFSET X/Y0.500

ENERGY: X2.000 / Y1.875. TOLERANCE: X6.000 / Y1.875.

NEGOTIATE: X4.000 / Y1.750. LED: X4.000 / Y2.375.

Window: X2.625 / Y3.125; 2.750 x 1.250, R0.03125.

Plate local window: X2.125 / Y2.625.

Four plate fixing slots: 0.125 x 0.1875, Y-oriented.

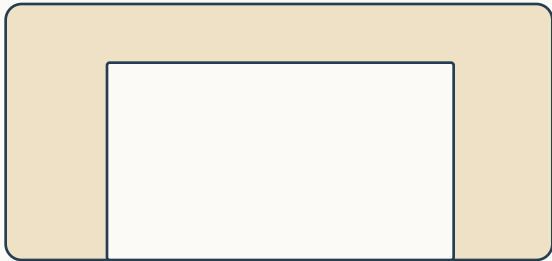
Slot centres local X0.125/6.875 and Y0.125/4.375.

Align the actual display active area, not the PCB centre. Nominal active area 66.89 x 29.05 mm; board 82 x 38 mm. Mounting pattern and connector clearance require the received part.

Carrier supports PCB mounting lands. The clear window bonds at its perimeter to the plate, with at least 0.040 air gap to display glass. Never clamp the glass or ribbon.

KNOB SECTION / SCHEMATIC

DIA 1.375 x 0.625



UNDERSIDE POCKET

DIA 0.625 x 0.500 DEEP

ROOF 0.125

TRIAL HUB FIT ONLY

Wood sleeve bonds to a bought T18 knob, then fits the pot shaft. Preserve full shaft engagement and 0.0625 minimum wood-to-plate clearance. Coupon-test the soft coating/bond and wobble before making both knobs.

Pot nuts sit on 1.5 mm metal, not through 0.500 wood. Engrave legends/pointers after measuring the actual travel and calibrating all three zones. Nominal 300-degree pot travel is not a generic 270-degree dial.



USB power through Pico only. Controls at 3.3 V. Display connects through its correctly aligned Pico header.

PIN ASSIGNMENTS

|                      |                         |
|----------------------|-------------------------|
| ENERGY / TOLERANCE   | GP26 pin31 / GP27 pin32 |
| NEGOTIATE / LED      | GP16 pin21 / GP17 pin22 |
| Display DC / CS      | GP8 pin11 / GP9 pin12   |
| Display SCK / MOSI   | GP10 pin14 / GP11 pin15 |
| Display RESET / BUSY | GP12 pin16 / GP13 pin17 |

Each pot wiper: 1k series, then 100k and 100 nF to AGND. Button: 10k pull-up, switch to ground. LED: 1k series. See wiring guide for supplies, polarity and programming details.

ONE PRESS, ONE CAPTURED PROPOSAL

1. Require button release and stable dial settings.
2. Freeze both levels and select an authored response.
3. Update once; discard presses while busy.
4. Sleep display; wait for allowed next refresh.
5. Fault on bounded timeout; do not retry endlessly.

Baseline: 180 seconds between full refreshes while panel is qualified, including initial boot wait. Faster partial updates are a later hardware test. The object must prove this pacing is enjoyable; change the display if it is not.

The screen retains the LAST PROPOSAL when unplugged. Footer shows the captured settings. A separate LED indicates ready/updating/wait/fault. Moving the dials alone never refreshes the screen.

01 / FIT AND SOURCING

A 2.500-inch aperture would clip the 66.89 mm active area. Increased it to 2.750 with a larger clear window. Rejected the retired Inky Pack route. Keep the selected Waveshare model conditional on stock, revision and driver checks.

02 / INTERACTION AND ELECTRICAL

The initial model exposed held-button repeats and mutable dial snapshots. Revised to fresh-edge triggering, immutable requests and fault timeout. Separate LED reports powered status while e-paper can retain stale text.

03 / COMPLETE PACKAGE

Recheck inch contours, cavity/control reserves, knob roof, window clearance and text bounds. Inspect all rendered pages. Verify no queued requests, all nine settings, and agreement between wiring, pseudocode and assembly notes.

FIRST BUILD: ONE DIAL + DISPLAY COUPON

Fit one purchased knob inside one wood coupon. Bench-test the display, button and both pots. Put the screen behind the clear-window coupon and read it at seated distance. Qualify refresh behaviour before making the enclosure.

This packet supplies prototype CAD and a tested policy model, not hardware-tested firmware. Component bores, hub adhesive, carrier mounts, USB notch and final refresh profile remain deliberate physical-sample holds.

Source links and quantities: [sourcing.md](#) | Wiring and initial logic: [wiring-and-pseudocode.md](#)

Editable text: [responses.json](#) | Review evidence: [review-history/](#) | Shop record: [fit-log.md](#)